**MATH2241 Introduction to Mathematical Analysis**
**Test 2**

November 25, 2022 Name:
12:30 pm - 1:30 pm Student Number:

*Attempt ALL Seven Questions*

Q1. (8 Marks) Let $g : \mathbb{R} \to \mathbb{R}$ be differentiable and satisfy $|g'(x)| \leq \frac{1}{2}$. Show that the function $f(x) = x - g(x)$ is injective.

Q2. (12 Marks)

(i) Why is the function $f(x) = |x|^{5/2}\cos(|x|^{3/2})$ continuously differentiable on $[-1, 1]$ ?

(ii) Why does $f$ have a minimum value on this interval ?

Q3. (22 Marks) Consider the function

$$f(x) = \frac{-x(x+1)(x-100)}{x^{44} + x^{34} + 1}$$

for $x \in [0, 100]$.

(i) Explain why $f$ has derivatives of all orders.

(ii) Compute $f'(0)$.

(iii) Show that there exists $\delta > 0$ such that $f(x) > 0$ for $0 < x < \delta$.

(iv) Show that there must exist a point $x$ with $f'(x) = 0$ and $0 < x < 100$.

Q4. (16 Marks)

(i) For a series $\{a_n\}$, if the series $\displaystyle\sum_{n=1}^{\infty} a_n$ converges, does it necessarily follow that for every subsequence $\{a_{n_k}\}$, the series $\displaystyle\sum_{k=1}^{\infty} a_{n_k}$ converges ?

(ii) If $\displaystyle\sum_{n=1}^{\infty} a_n$ converges absolutely, does $\displaystyle\sum_{k=1}^{\infty} a_{n_k}$ converge absolutely?

Q5. (16 Marks) If a series of nonnegative real numbers $\displaystyle\sum_{n=1}^{\infty} a_n$ converges, prove that also the series $\displaystyle\sum_{n=1}^{\infty} a_n^2$ converges. Does the series $\displaystyle\sum_{n=1}^{\infty} \sqrt{a_n}$ necessarily converge?

Q6. (14 Marks)

(i) Prove or disprove (with counterexample) that a product of two uniformly continuous functions is uniformly continuous.

(ii) Let $f : \mathbb{R} \to \mathbb{R}$ with $f(x) = e^x$. Is $f$ uniformly continuous (state the reason) ?

Q7. (12 Marks) Compute the limit (showing the steps)

(i)

$$\lim_{x \to 0} \frac{e^{-1/x^2}}{x^3};$$

(ii)

$$\lim_{x \to \infty} x \sin x.$$

Q1. MVT: $f(x) - f(y) = (x - y)f'(t)$, $f'(t) = 1 - g'(t) \neq 0$ so $f$ is 1-1.

Q2. $|x|^{3/2}$ is differentiable on $[-1, 1]$ We know cos is differentiable on $[-1, 1]$ so $\cos(|x|^{3/2})$ is differentiable on $[-1, 1]$ as composite of differentiable functions on $[-1, 1]$. $f$ is continuous on $[-1, 1]$, $f$ attains a minimum value.

Q3. (i) Polynomials are infinitely differentiable and the quotient of two infinitely differentiable functions is infinitely differentiable on any interval on which $q \neq 0$. Since $x^{44} + x^{34} + 1 > 0$ for $x \in \mathbb{R}$, it follows that $f = p/q$ is infinitely differentiable on $\mathbb{R}$. (ii) By quotient rule, $f'(0) = 100$. (iii) Since $f'(x)$ is continuous, there exists $\delta > 0$ such that $f'(x) > 0$ if $x \in [0, \delta)$. By the mean valued theorem for $x \in (0, \delta)$, $f(x) = xf'(y)$, $y \in (0, \delta)$, it implies that $f(x) > 0$. (iv) From the form of $f$, $f(100) = 0$ so, again by the mean value theorem $f(100) - f(0) = 0 = 100$f'(x) for some $x \in (0, 100)$.

Q4. First, let $\{a_n\}$ be a sequence such that the series $\sum_{n=1}^{\infty} a_n$ that converges absolutely. Let $\{a_{n_k}\}$ be any subsequence. Define $\{b_n\}$ to be the sequence such that $b_n$ equals $a_n$ if $n$ equals $n_k$ for some $k \in \mathbb{N}$, or else $b_n$ equals 0. Then for every $n \in \mathbb{N}$, $|b_n|$ is at most $|a_n|$. Thus, by the Comparison Test, also the series $\sum_{n=1}^{\infty} b_n$ converges absolutely. Of course $\sum_{n=1}^{\infty} b_n$ is the same as the series $\sum_{k=1}^{\infty} a_{n_k}$, hence it converges absolutely.

On the other hand, for a series $\sum_{n=1}^{\infty} a_n$ that converges conditionally, for a subsequence, it need not converge. For instance, for the alternating harmonic series $\{(-1)^n/n\}$, the series does converge conditionally. Yet for the subsequence $\{a_2, a_4, a_6, \ldots\}$, the associated series is $2\sum_{k=1}^{\infty} 1/k$, and this diverges.

Q5. Since the series $\sum_{n=1}^{\infty} a_n$ converges, by the Cauchy criterion, for every $\epsilon > 0$ there exists an integer $N \in \mathbb{N}$ such that for every pair of integers $m, l$ with $l \geq m \geq N$, $|\sum_{n=m}^{l} a_n|$ is less than $\epsilon$. In particular, taking $\epsilon = 1$ and taking $l = m$, there exists an integer $N \in \mathbb{N}$ such that for every $m \geq N$, $0 \leq a_m < 1$. Thus, by the order axioms, also $0 \leq a_m^2 < a_m < 1$. Therefore the series $\sum_{n=N}^{\infty} a_n^2$ is bounded by the convergent series $\sum_{n=N}^{\infty} a_n$. By the Comparison Test, the series $\sum_{n=N}^{\infty} a_n^2$ is convergent. Since the tail of the series is

convergent, the entire series $\sum_{n=1}^{\infty} a_n^2$ is convergent.

For a convergent series $\sum_{n=1}^{\infty} a_n$, the series $\sum_{n=1}^{\infty} \sqrt{a_n}$ need not converge. One example is the series: $a_n = \frac{1}{n^2}$ but the harmonic series diverges.

Q6. (i) False. Consider $f, g : \mathbb{R} \to \mathbb{R}$ with $f(x) = x, g(x) = x$. (ii) $f$ is not uniformly continuous. Suppose to the contrary that $f$ is uniformly continuous. Take $\epsilon = 1$, for any $\delta > 0$ let $a = \delta/2$. As we have $\lim_{x\to\infty} e^x(e^a - 1) = \infty$, there is $x_0 \in \mathbb{R}$ such that $e^{x_0}(e^a - 1) > 1$. Letting $y_0 = x_0 + a$, one has $|x_0 - y_0| < \delta$ while $|e^{x_0} - e^{y_0}| = e^{x_0}|e^a - 1| > 1$, a contradiction.

Q7.

$$\lim_{x\to 0} \frac{e^{-1/x^2}}{x^3} = 0$$

via L.H. rule calculation. Consider the sequence $x_n = \pi/2 + 2n\pi$. For some $N \in \mathbb{N}$, $\pi/2 + 2N\pi > a$ so that for all $n > N$, $\pi/2 + 2n\pi > a$, but $f(\pi/2 + 2n\pi) = \pi/2 + 2n\pi > a$. Hence there is no real number $L$ such that $\lim_{x\to\infty} x \sin x$ does not exist.